Stability map of few-body configurations: Routh, Maxwell, hierarchy, and dissipation

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We verify whether a simple symplectic integrator reproduces, without prior information, the classical stability criteria for few-body configurations. The Gascheau-Routh limit for the Lagrange triangle ($\mu \approx 0.0198$ for masses $1{:}\mu{:}\mu$), the Maxwell criterion for rings with a central body ($n \geq 7$ and $\mu < 2.298/n^3$), and the hierarchical criterion of Mardling and Aarseth are reproduced, the last with good agreement for $m_3 \geq 1$. Within the linearly stable region, we document islands of instability associated with the 2:1 resonance between the eigenfrequencies, at $\beta \approx 0.0237$ to 0.0239, in agreement with Deprit and Deprit-Bartholomé (1967). A linear drag destroys the gyroscopic stability of the triangle in a time approximately proportional to $1/\gamma$, as predicted by the Kelvin-Tait-Chetaev theorem.

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I. INTRODUCTION

Before proposing new stabilization mechanisms, it is advisable to verify whether the simulator reproduces the classical criteria. This note reports that verification for three families of configurations: the Lagrange triangle with unequal masses [1, 2], rings with a dominant central body [4], and hierarchical systems [6]. It adds the effect of dissipation on gyroscopic stability [7] and documents islands of resonant instability within the linearly stable region [3]. The simulation receives no information about the criteria; the thresholds emerge from direct integration.

II. METHOD

Units with $G = 1$; leapfrog integration with a fixed step of 0.002, or an adaptive step proportional to $r_{\min}^{3/2}$ in cases with close encounters; five perturbations per configuration, from 10^{-4} to 10^{-3} in position or velocity; integration limit between 600 and 3000 time units, depending on the family. A configuration is recorded as stable when no perturbation disrupts it within the limit, according to explicit geometric criteria for each family: relative deformation of the sides above 30%, angular deviation between neighbors above 30% of the spacing, or disruption of the inner binary.

III. RESULTS

A. Lagrange triangle: the Gascheau-Routh limit

For masses m_1, m_2, m_3 in a rigidly rotating equilateral configuration, linear stability requires [1, 2]

$$(m_1 + m_2 + m_3)^2 > 27\,(m_1 m_2 + m_2 m_3 + m_3 m_1). \qquad (1)$$

With masses $1 : \mu : \mu$, the limit is $\mu \approx 0.0198$; that is, the dominant body must have more than about 50 times the mass of each of the others. The simulation reproduces the limit: all perturbations remain bounded up to $\mu = 0.0195$, with a maximum deformation of 3.9%; at $\mu = 0.0205$ the deformation grows slowly to 18.5% without disruption within the integrated interval; from $\mu = 0.022$ onward all perturbations disrupt the triangle, with a median survival time that falls from about 60 time units at $\mu = 0.022$ to about 8 at $\mu = 1$.

B. Islands of resonant instability

The Routh condition is linear. Within the linearly stable region, the two eigenfrequencies of small oscillations about the equilateral configuration, in the rotating frame, are given by

$$s^2 = \frac{1 \pm \sqrt{1 - 27\beta}}{2}, \qquad \beta = \frac{m_1 m_2 + m_2 m_3 + m_3 m_1}{(m_1 + m_2 + m_3)^2}. \qquad (2)$$

When the ratio s_1/s_2 is a small integer, nonlinear terms can transfer energy between the modes. The scan with masses $1 : \mu : \mu$ found exactly this: at $\mu = 0.0122$ and $\mu = 0.0124$, where $s_1/s_2 = 2.02$ and 1.996, the larger perturbations (10^{-3}) disrupted the triangle, whereas at $\mu = 0.0118$ and $\mu = 0.0126$, with ratios 2.075 and 1.970, all perturbations remained bounded. With masses $1 : \mu : 0.25\mu$, the same 2:1 resonance appeared at $\mu = 0.020$ ($\beta \approx 0.0239$), within a range that the Routh criterion classifies as stable up to $\mu \approx 0.032$.

The result is consistent with the classical nonlinear analysis of the restricted problem, in which the 2:1 resonance between the libration frequencies occurs at $\mu \approx 0.0243$ [3]; in terms of β, the simulated value ($\beta \approx 0.0237$ to 0.0239) coincides. The 3:1 resonance, predicted near $\mu \approx 0.0068$ in this family, did not manifest itself at the amplitudes tested.

Table I. Survival time, in time units, of rings of n bodies with a central body, for three perturbations, at multiples of the Maxwell mass limit. "Stable" indicates survival under all perturbations up to $t = 600$.

n	$0.1\times$ limit	$0.5\times$	$0.9\times$	$1.5\times$
3	34 to 46	19 to 23	16 to 19	13 to 16
4	58 to 70	30 to 35	23 to 28	20 to 23
5	80 to 102	45 to 56	36 to 43	29 to 36
6	122 to 159	69 to 87	58 to 73	34 to 40
7	stable	stable	stable	32 to 37
8	stable	stable	stable	32 to 37
9	stable	stable	stable	29 to 36

Table II. Critical distance of the third body, in units of the inner binary separation, obtained by simulation and from the Mardling-Aarseth criterion.

m_3	Simulated critical D	Mardling and Aarseth
0.1	2.4	2.86
1	3.1	3.29
10	5.8	5.73

C. Rings with a central body: the Maxwell criterion

In his 1859 essay on the rings of Saturn, Maxwell showed that n equal masses μ, equally spaced in a circular orbit around a dominant body, can be stable only if $n \geq 7$, and provided that μ remains below approximately $2.298/n^3$ times the central mass [4]. The simulation yielded the times in Table I.

The transition between $n = 6$ and $n = 7$ is sharp, and finer scans placed the mass limit between 1.0 and 1.1 times the Maxwell value for $n = 7$, 8, and 10. For $n \leq 6$, the survival time increases as the ring mass decreases, but the instability persists, in agreement with the linear stability analysis of configurations with a dominant mass [5].

D. Hierarchical systems

With a circular inner binary of equal masses and unit separation, and a third body of mass m_3 in a circular outer orbit at distance D, the simulated stability threshold is compared, in Table II, with the empirical criterion of Mardling and Aarseth [6], which, for circular and coplanar orbits, reduces to

$$\frac{D}{a_{\rm in}} > 2.8\left(1+\frac{m_3}{m_1+m_2}\right)^{2/5}. \qquad (3)$$

The agreement is good for $m_3 \geq 1$. For $m_3 = 0.1$, the simulated threshold is lower, which is expected: the criterion was calibrated to be conservative and covers eccentric orbits, and the integration here is limited to about 340 periods of the inner binary.

E. Dissipation and gyroscopic stability

The triangular Lagrange points are maxima of the effective potential in the rotating frame. Their stability, when it exists, is gyroscopic: the Coriolis force converts the displacement into motion around the point. The Kelvin-Tait-Chetaev theorem states that the gyroscopic stability of a potential maximum does not survive dissipation [7]. To verify this, a linear drag $-\gamma(\mathbf{v}-\Omega\times\mathbf{r})$ was applied to the smaller bodies of a stable triangle with $\mu = 0.01$. Table III summarizes the result.

The disruption time scales approximately as $1/\gamma$, as expected for a dissipation-induced secular instability. For $\gamma = 10^{-4}$, the characteristic time exceeds the integrated interval.

IV. DISCUSSION

In all stable families, the system decomposes into approximately Keplerian subsystems, and the coupling between them remains below a threshold. The separation of scales comes from mass (Routh), from distance (hierarchy), or from number and symmetry with a dominant mass (Maxwell). Within the stable regions, resonances between eigenfrequencies open islands of instability, and dissipation destroys stability of gyroscopic origin.

The main limitations are the planar geometry, circular orbits, the small number of perturbations per point, and finite integration times, which differ between families. The thresholds should be read as reproductions of known criteria, which validate the simulator, and not as new results. The exception is the numerical location of the 2:1 resonant island in the $1 : \mu : 0.25\mu$ family, which we did not find tabulated in the literature consulted.

V. CONCLUSIONS

The simulator reproduces, without prior knowledge of them, the criteria of Gascheau and Routh, of Maxwell, and of Mardling and Aarseth, and confirms the Kelvin-Tait-Chetaev theorem. The observed regularity suggests a common heuristic, namely that stability depends on a small coupling between Keplerian subsystems. The next note in the series tests whether these thresholds can be described by a single coupling number.

TRANSPARENCY AND USE OF ARTIFICIAL INTELLIGENCE

The author is a lawyer and not a physicist. This work results from independent research, conducted with the assistance of artificial intelligence (a language model) in the literature searches, the programming of the simulations, the analysis of the results, the writing, and the translation. The author formulated the questions, followed the experiments, and is responsible for the published content.

Table III. Survival of the stable Lagrange triangle ($\mu = 0.01$) under linear drag with coefficient γ, for two perturbations.

γ	Survival
0	stable up to $t = 3000$
10^{-4}	stable up to $t = 3000$
10^{-3}	1459 and 2269
10^{-2}	162 and 231

DATA AND CODE AVAILABILITY

All numbers were obtained with scripts published at https://fisica-revisitada.vercel.app/en/code/, which run on Node.js without external dependencies and allow the results to be reproduced.

CONFLICT OF INTEREST

The author declares no conflicts of interest. This work received no funding. This text has not undergone peer review and should be read as a working note. Corrections are recorded in the note itself, with the date.

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[1] J. Gascheau, Examen d'une classe d'équations différentielles et application à un cas particulier du problème des trois corps, C. R. Acad. Sci. Paris 16, 393 (1843).

[2] E. J. Routh, On Laplace's three particles, with a supplement on the stability of steady motion, Proc. London Math. Soc. 6, 86 (1875).

[3] A. Deprit and A. Deprit-Bartholomé, Stability of the triangular Lagrangian points, Astron. J. 72, 173 (1967). doi.org/10.1086/110213

[4] J. C. Maxwell, On the Stability of the Motion of Saturn's Rings (Macmillan, Cambridge, 1859).

[5] R. Moeckel, Linear stability of relative equilibria with a dominant mass, J. Dyn. Differ. Equ. 6, 37 (1994). doi.org/10.1007/BF02219187

[6] R. A. Mardling and S. J. Aarseth, Tidal interactions in star cluster simulations, Mon. Not. R. Astron. Soc. 321, 398 (2001). doi.org/10.1046/j.1365-8711.2001.03974.x

[7] N. G. Chetaev, The Stability of Motion (Pergamon, Oxford, 1961); W. Thomson and P. G. Tait, Treatise on Natural Philosophy (Cambridge University Press, 1879).